Project Report On

# G-Talk (Real-Time Chat Application)

Submitted in partial fulfilment for the award of

**Post Graduate Diploma in Advanced Computing**

**From C-DAC ACTS (Pune)**

Guided by

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Centre of Development of Advanced Computing(C-DAC), Pune

# CERTIFICATE

TO WHOMSOEVER IT MAY CONCERN

**This is to certify that**

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have successfully completed their project titled

**"G-Talk (Real Chat Application)"**

**Under the Guidance of**

**Mr. Pawan Khatri**

**Project Guide** **HOD ACTS**

# ACKNOWLEDGEMENT

This project **"GTalk – A Discord-like Web Application"** was a great learning experience for us, and we are submitting this work to **Advanced Computing Training School (C-DAC ACTS)**.

We are very glad to mention the name of **Mr. Pawan Khatri** for his valuable guidance throughout the development of this project. His continuous support, technical guidance, and constructive feedback helped us overcome various obstacles and intricacies encountered during the course of project work.

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# 1. INTRODUCTION

GTalk is a modern, Discord-like real-time communication web application designed to facilitate seamless interaction and collaboration among users. The platform enables users to communicate through real-time text messaging within organized servers and channels, providing an engaging and user-friendly environment suitable for communities, teams, and groups. GTalk focuses on delivering fast, reliable, and secure communication with a clean and intuitive interface.

Powered by a combination of robust and scalable technologies, GTalk redefines the way users interact in online communication platforms. The backend of the application is developed using **Spring Boot**, exposing secure and efficient **RESTful APIs** that handle user authentication, authorization, and core business logic. **JWT (JSON Web Token)**–based authentication is implemented to ensure secure access control and session management across the application.

To enable real-time messaging functionality, **WebSocket technology** is utilized, allowing instant message delivery with low latency. This ensures smooth, bidirectional communication between clients and the server, providing a chat experience comparable to modern messaging platforms. **Redis** is integrated to enhance performance through fast in-memory data caching and efficient handling of frequently accessed data, improving overall system responsiveness.

GTalk uses **MySQL** as its primary relational database to store user details, server information, channel metadata, and message records in a structured and reliable manner. For efficient and scalable media storage, **MinIO** is incorporated to manage file uploads such as images and shared documents, ensuring secure and high-performance object storage.

The frontend of the application is developed using **React.js**, offering a dynamic and interactive user interface. React's component-based architecture enables efficient state management and real-time UI updates, providing a smooth and responsive user experience. Features such as user registration, login, server

creation, channel navigation, and real-time chat are seamlessly integrated into the interface.

GTalk is designed with a **responsive layout**, ensuring optimal usability across different devices and screen sizes. The application adapts dynamically to various resolutions, offering intuitive navigation, touch-friendly components, and fast loading times. By combining real-time communication, secure authentication, scalable storage, and a modern user interface, GTalk delivers a reliable, efficient, and user-centric communication platform suited for contemporary web-based collaboration.

# 2. SOFTWARE AND HARDWARE REQUIREMENTS

**Server:**

Processor: Intel Core i5 or equivalent AMD processor.

RAM: Minimum 8GB RAM.

Storage: SSD storage for improved performance.

Network: Ethernet or Wi-Fi connectivity.

Operating System: Linux distribution (Ubuntu, CentOS) preferred for server deployment, Windows.

**Client Devices:**

Processor: Dual-core processor or higher.

RAM: Minimum 4GB RAM.

Storage: Sufficient storage for caching and local data.

Network: Ethernet or Wi-Fi connectivity.

Browser: Compatible with latest versions of popular browsers like Google Chrome, Mozilla Firefox, and Safari.

# 3. TOOLS AND TECHNOLOGIES USED

- Spring Boot
- Spring Data JPA
- RESTful Web Services
- WebSocket
- Spring Web
- MYSQL Database
- JWT (JSON Web Token)
- Flyway
- Spring Security
- React JS
- HTML and CSS
- MinIO/S3 Object Storage
- Redis/Elastic Cache Database
- Docker

1. Spring Boot: Spring Boot is used to develop the backend of the GTalk application. It provides a robust and scalable framework for building enterprise-level Java web applications. Spring Boot simplifies configuration, dependency management, and application deployment, enabling rapid backend development.

2. Spring Data JPA: Spring Data JPA is implemented for database interaction, enabling seamless communication with the MySQL database. It simplifies data persistence and retrieval by reducing boilerplate code and providing repository-based access to relational data.

3. RESTful Web Services: GTalk utilizes RESTful web services to enable communication between the frontend and backend components. These services follow REST principles, ensuring stateless, standardized, and secure API communication for operations such as user authentication, server management, and channel handling.

4. WebSocket: WebSocket technology is used to enable real-time messaging functionality in GTalk. It allows full-duplex communication between the client and server, ensuring instant message delivery with low latency, which is essential for a chat-based communication platform.

5. Spring Web: Spring Web is used to handle HTTP requests and responses, manage controllers, and expose REST APIs. It plays a vital role in routing client requests and coordinating backend services efficiently.

6. MySQL Database: MySQL is used as the primary relational database to store structured data such as user profiles, server details, channel metadata, and message information. It provides reliability, data integrity, and efficient querying capabilities.

7. Redis Cache Database: Redis is integrated as an in-memory cache database to enhance application performance. It is used for fast data access, caching frequently used information, and improving the responsiveness of real-time operations.

8. JWT (JSON Web Tokens): JWT is implemented for secure user authentication and authorization. It enables stateless authentication by securely transmitting user identity information between the client and server, ensuring protected access to application resources.

9. MinIO Object Storage: MinIO is used for storing and managing user-uploaded files such as images and shared documents. It offers scalable, secure, and high-performance object storage, making it suitable for handling media content in GTalk.

10. Spring Security: Spring Security is used to implement authentication and authorization mechanisms within the application. It integrates seamlessly with JWT to protect REST APIs and restrict access based on user roles.

11. React JS: React JS is used to develop the frontend of the GTalk application. It provides a component-based architecture that enables dynamic rendering, efficient state management, and real-time UI updates for a smooth user experience.

12. HTML and CSS: HTML and CSS are used to structure and style the frontend interface. They help in creating a responsive, visually appealing, and user-friendly design.

13. Docker: Docker is used to containerize the GTalk application, ensuring consistent runtime environments across development, testing, and deployment. It simplifies application deployment, improves scalability, and enhances portability.

## 4. PROJECT FLOW DIAGRAM

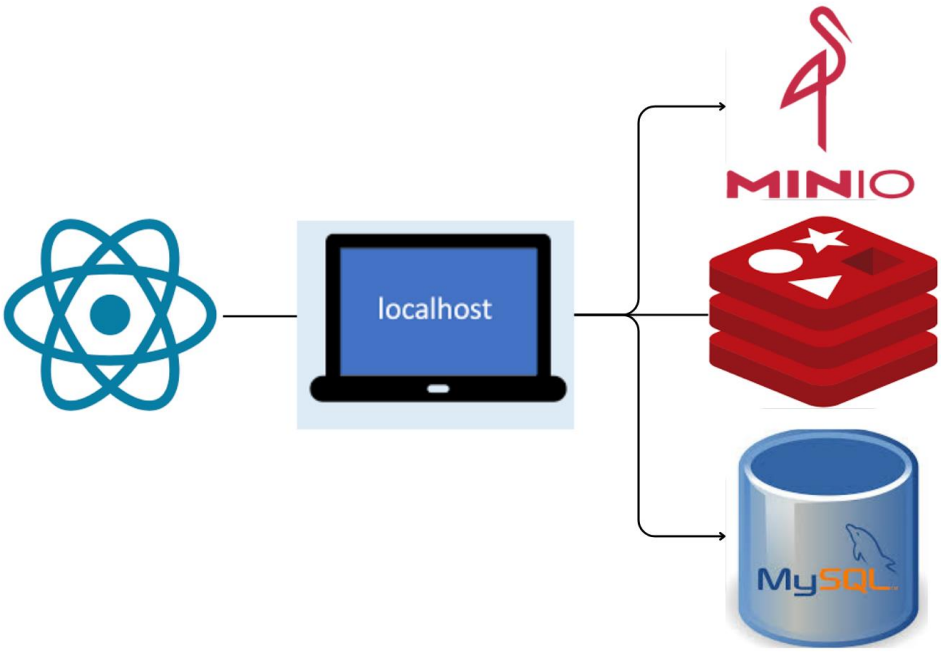


**Fig 4. 1 Local Development Stack**

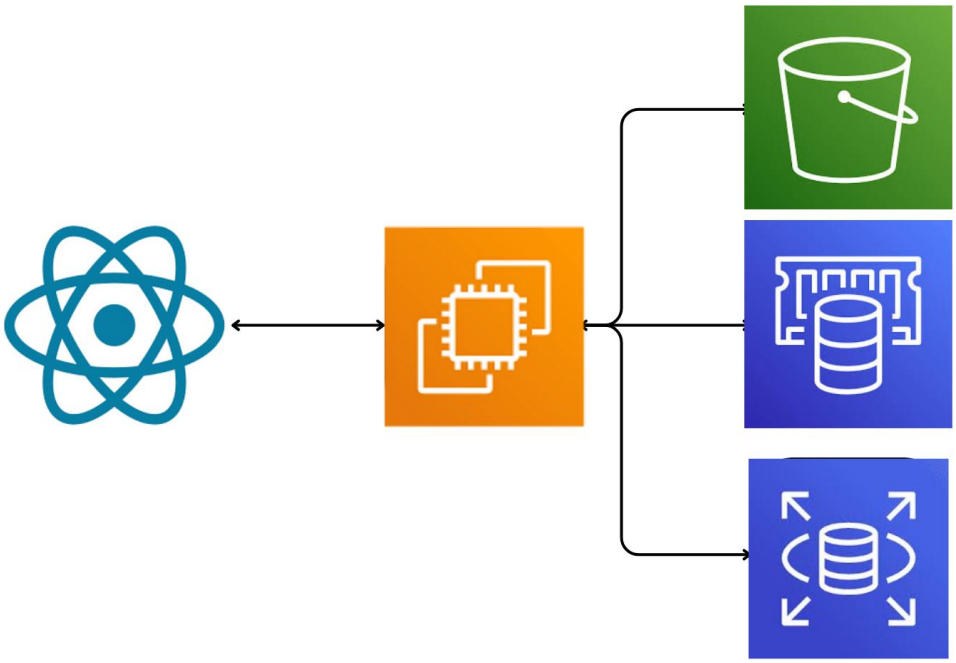

**Fig 4.2 Cloud Deployment Stack (AWS)**

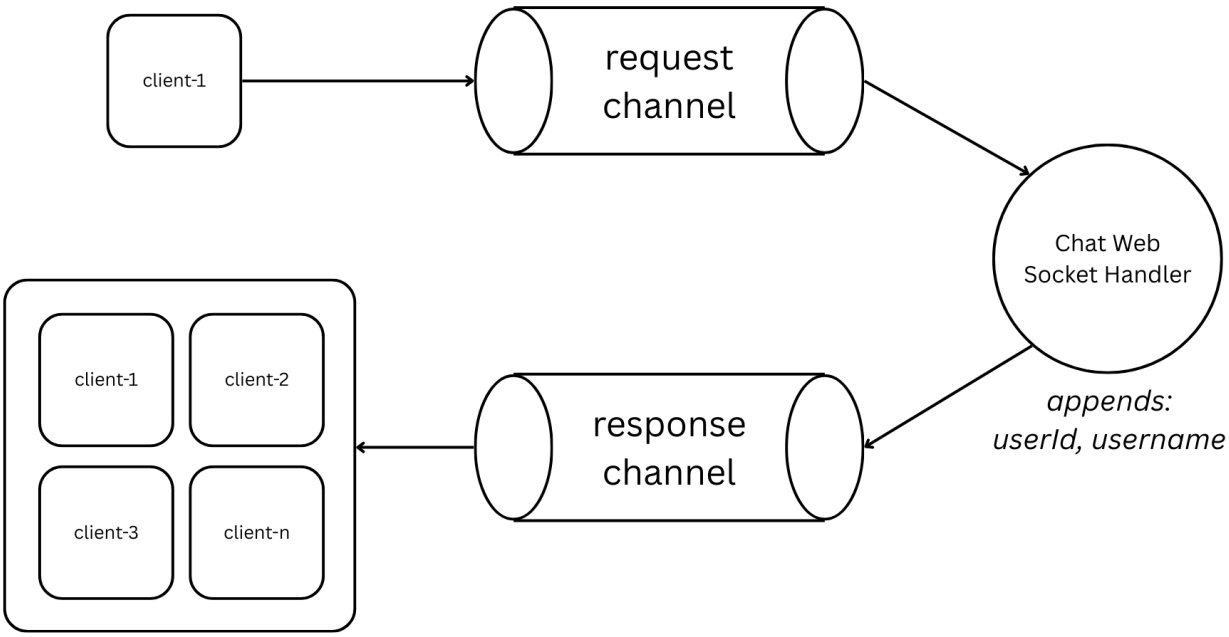


**Fig 4.3 WebSocket**

```
{
  "type": " ",
  "channelId": ,
  "payload": {
    "content": " ",
    "attachmentIds": []
  }
}
```

WsMessage Request

```
{
  "type": " ",
  "channelId": ,
  "payload": {
    "content": " ",
    "attachmentIds": []
  },
  "userId": ,
  "username": " "
}
```

WsMessage Response

**Fig 4.4 WebSocket Message Payload**

# 5. ER DIAGRAM

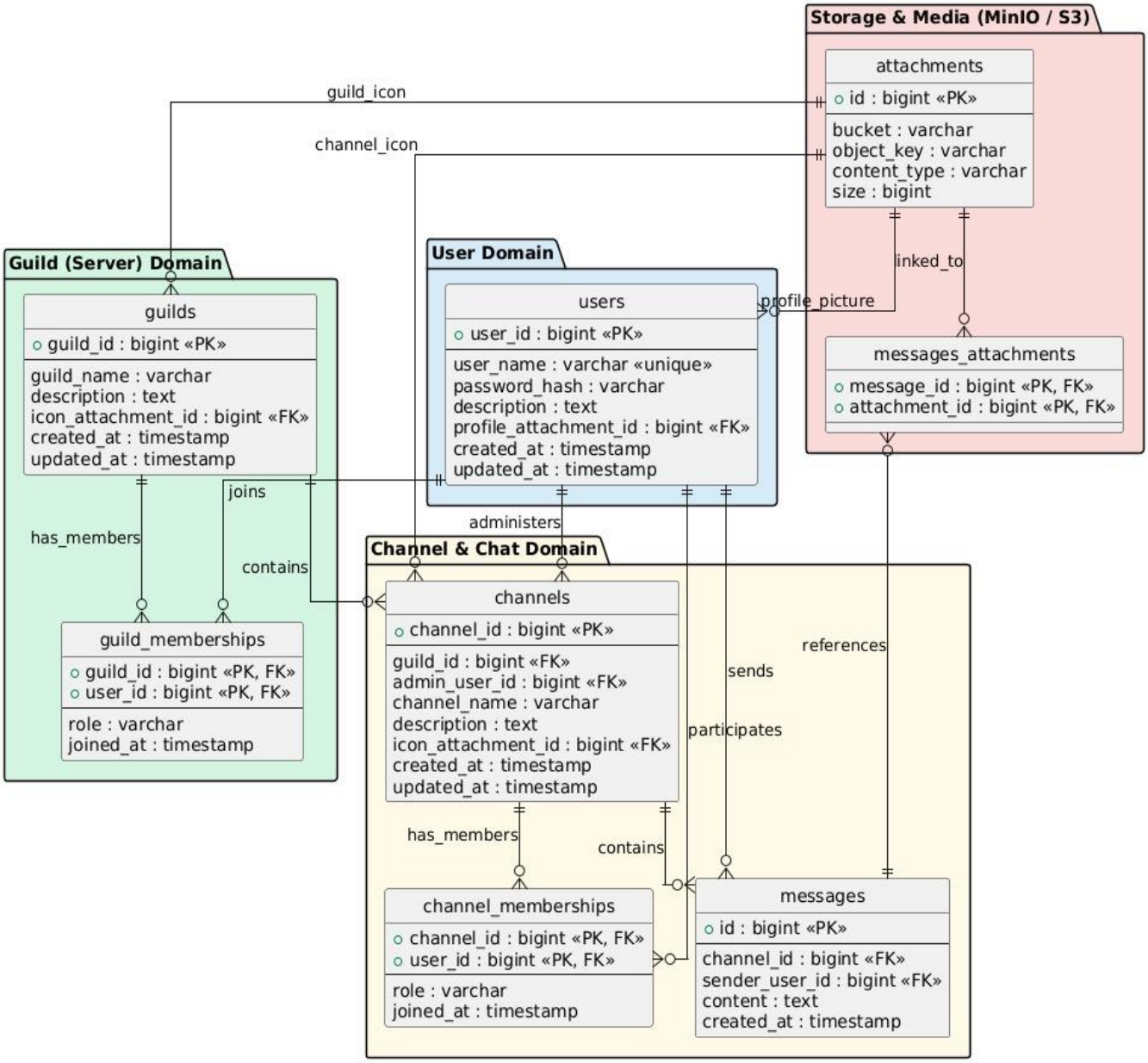


**Fig 5.1 Entity-Relationship Diagram**

# 6. USE CASE DIAGRAM

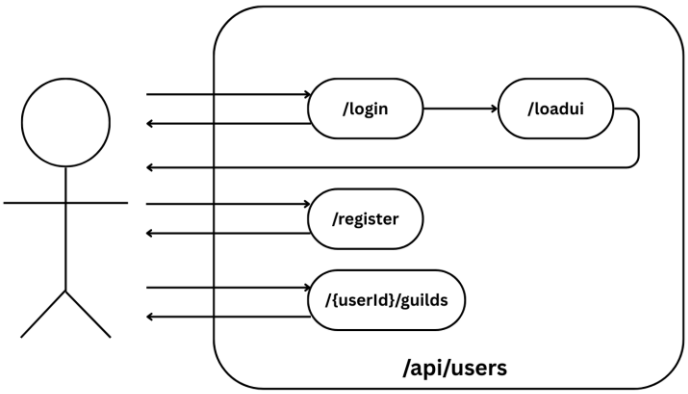


**Fig 6.1 User Controller**

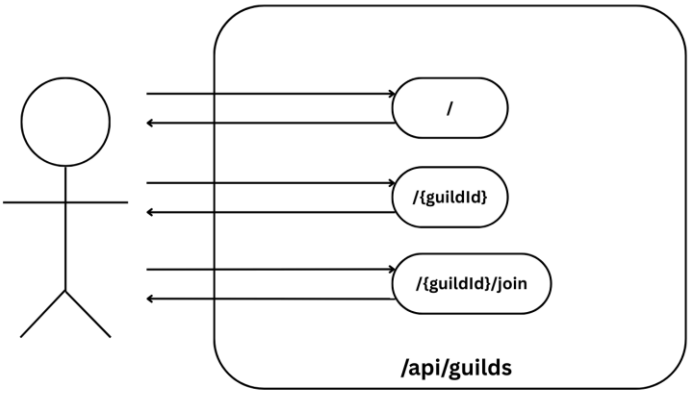


**Fig 6.2 Guild Controller**

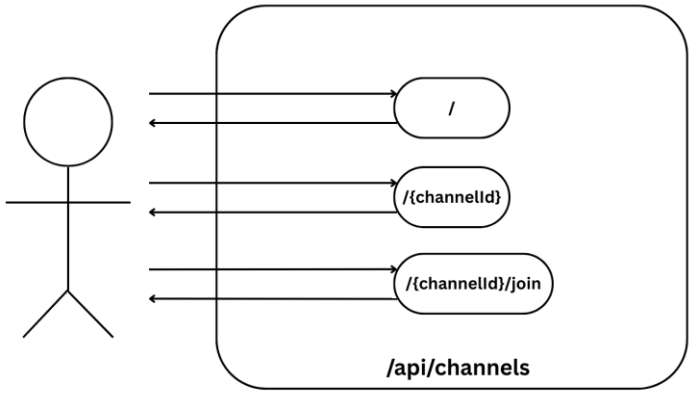


**Fig 6.3 Channel Controller**

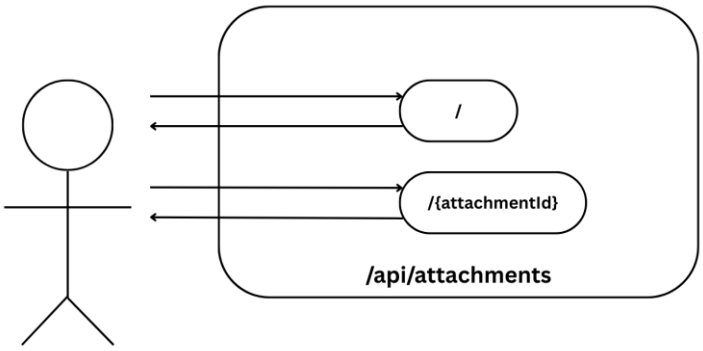


**Fig 6.4 Attachment Controller**

# 7. ADVANTAGE

## Use of MySQL Database

The GTalk project uses **MySQL** as its primary relational database for managing structured data such as user information, server details, channels, and message records. The advantages of using MySQL

include:

- **Reliability:** MySQL provides stable and consistent data storage, ensuring reliable access to application data.
- **Scalability:** MySQL supports horizontal and vertical scaling, allowing the application to grow with increasing users and data.
- **Data Integrity:** ACID compliance ensures data accuracy and consistency across transactions.
- **Security:** MySQL supports authentication, access control, and encryption to protect sensitive user data.
- **Wide Compatibility:** MySQL integrates seamlessly with Spring Boot and standard database APIs.
- **Cost-Effective:** Being open-source, MySQL reduces infrastructure and licensing costs.

## Use of Redis Cache Database

Redis is integrated into GTalk to enhance performance and responsiveness:

- **High Performance:** Redis provides in-memory data storage, enabling ultra-fast data access.
- **Reduced Database Load:** Frequently accessed data is cached, reducing the number of database queries.
- **Improved Real-Time Experience:** Redis helps in managing session data and frequently used metadata efficiently.
- **Scalability:** Redis supports distributed caching, making it suitable for large-scale real-time applications.

## Use of WebSocket for Real-Time Messaging

WebSocket plays a crucial role in enabling instant communication in GTalk:

- **Low Latency Communication:** Enables real-time, bidirectional communication between client and server.
- **Efficient Message Delivery:** Eliminates repeated HTTP requests, improving performance.
- **Scalable Architecture:** Suitable for chat applications with multiple concurrent users.
- **Enhanced User Experience:** Provides smooth and instant message updates without page refresh.

## Use of JWT for Authorization

JWT is implemented to ensure secure and efficient authentication in GTalk:

- **Stateless Authentication:** No server-side session storage is required, improving scalability.
- **Enhanced Security:** Tokens are digitally signed, ensuring data integrity and preventing tampering.
- **Reduced Server Load:** Authentication does not require database lookups for every request.
- **Cross-Platform Compatibility:** Easily integrates with web and mobile clients via HTTP headers.

## Use of MinIO and Amazon S3 for Object Storage

GTalk uses **MinIO** and **Amazon S3** to manage media and file storage:

- **Scalable Storage:** Handles large volumes of files efficiently.
- **High Availability:** Ensures reliable access to uploaded content.
- **Secure Data Storage:** Supports access control and encryption.
- **Cloud Compatibility:** MinIO's S3-compatible API enables seamless integration with cloud environments.

# 8. SCREENSHOTS

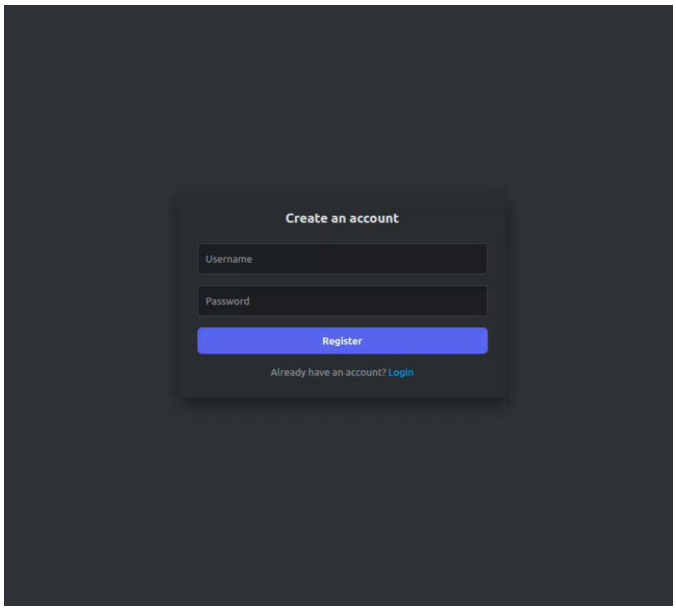


Fig 8.1 Register

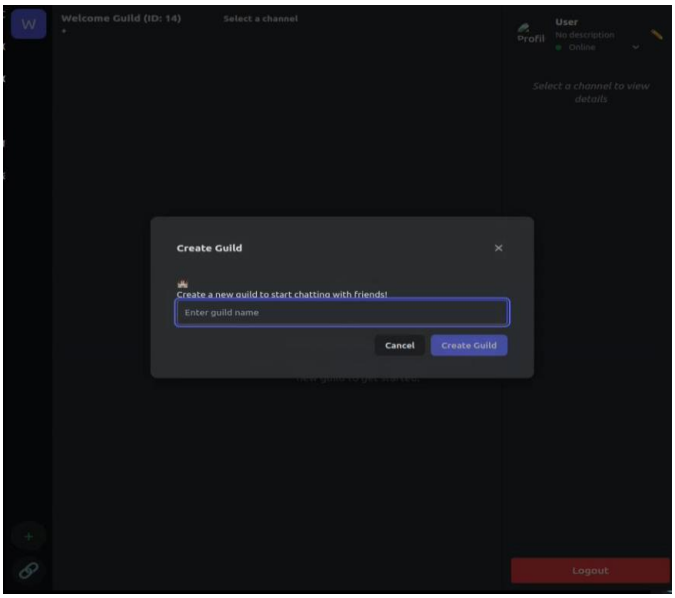


Fig 8.2 Create Guild

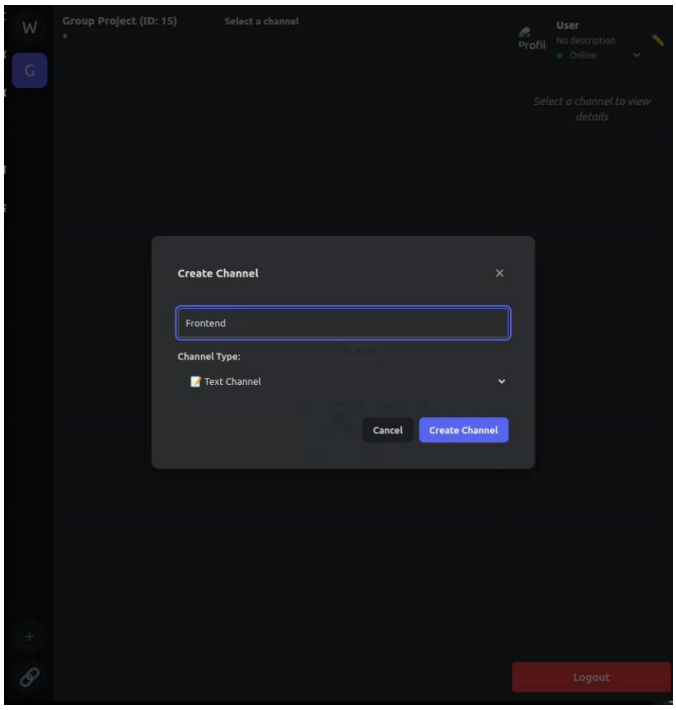


Fig 8.3 Create Channel

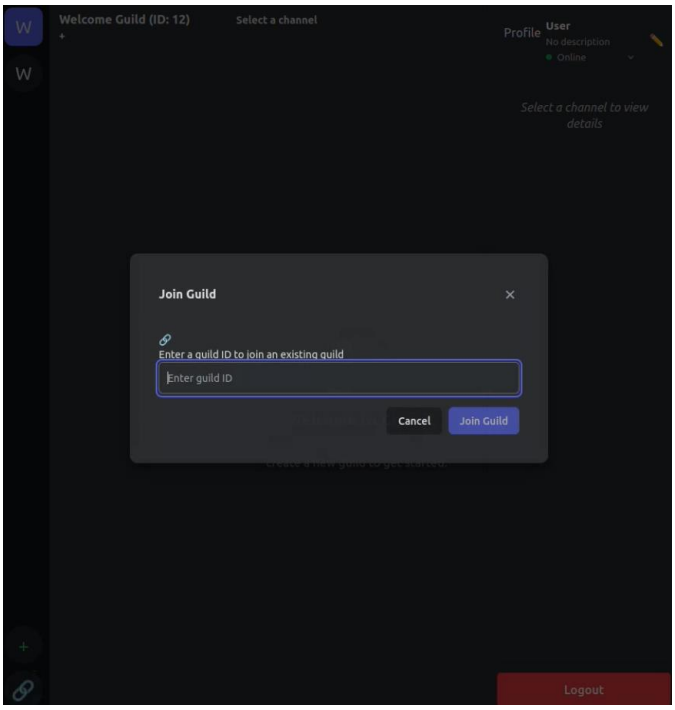


Fig 8.4 Join Guild

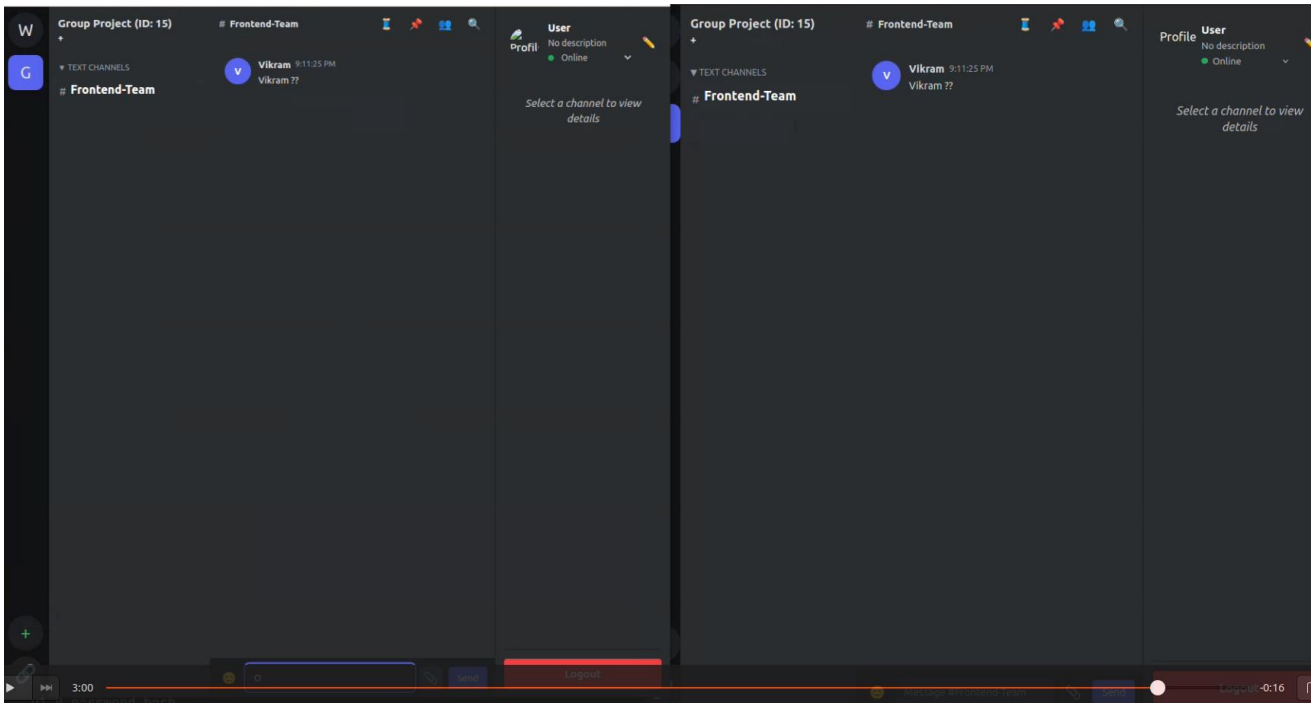


**Fig 8.4 Chat (1)**

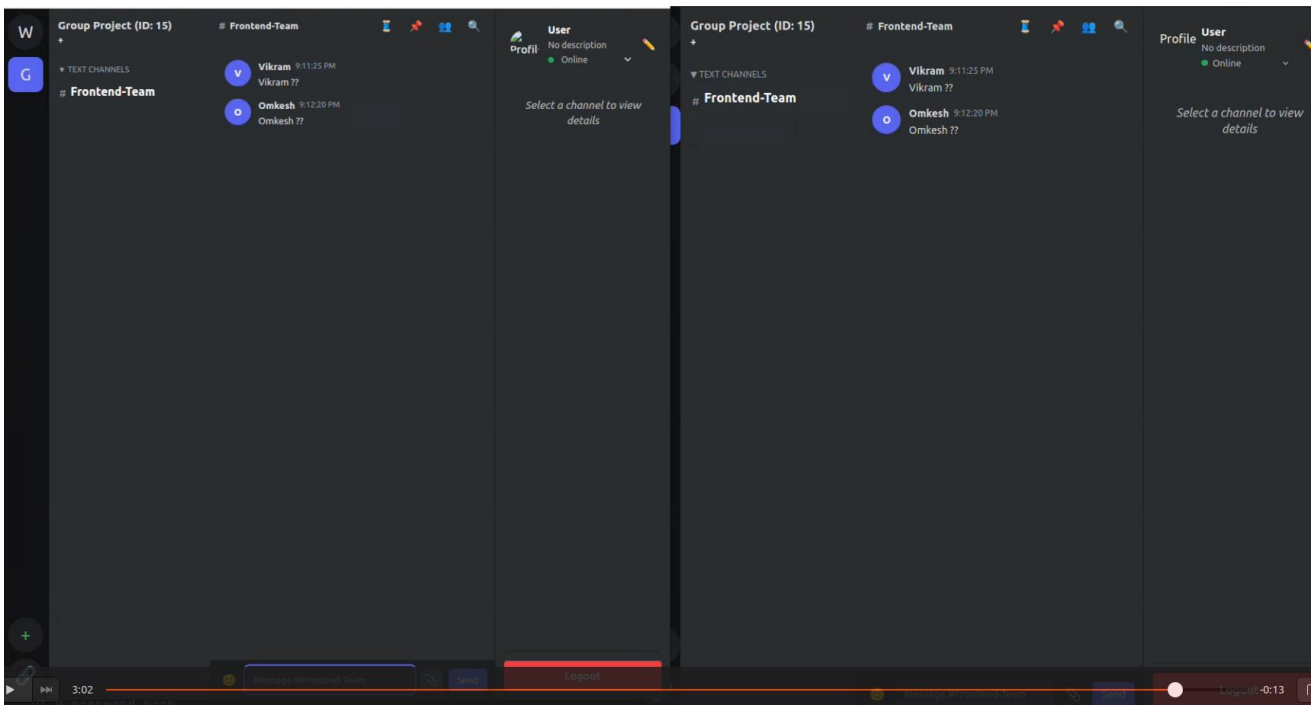


**Fig 8.5 Chat (2)**

# 9. FUTURE SCOPE

- **Subscription based model**:

  GTalk can introduce a subscription-based model offering users premium features such as enhanced storage limits, advanced customization options, priority access to new features, and exclusive server functionalities. Tiered subscription plans can be implemented to cater to different user requirements and engagement levels.

- **Voice and Video Communication**:

  Future enhancements can include the integration of **voice and video calling features**, enabling real-time audio and video communication within servers and private channels. This will make GTalk a more comprehensive collaboration platform similar to modern communication tools.

- **Notifications and Alerts:**

  Real-time push notifications for messages, mentions, and server activities can be added to ensure users stay informed even when they are not actively using the application.

# 10. CONCLUSION

In conclusion, the **"GTalk – A Discord-like Web Application"** project successfully integrates a range of modern and cutting-edge technologies to deliver a reliable, scalable, and real-time communication platform. By leveraging **Spring Boot**, **Spring Data JPA**, and **RESTful Web Services**, the application ensures robust backend functionality, secure user authentication, and efficient data management.

The GTalk project demonstrates a user-centric approach to online communication by incorporating **WebSocket-based real-time messaging**, **JWT-based authentication**, and a **React.js-powered frontend** for a smooth and interactive user experience. The integration of **MySQL** for structured data storage, **Redis** for caching and performance optimization, and **MinIO/Amazon S3** for scalable object storage enhances the overall efficiency and reliability of the system.

Overall, the GTalk project reflects a strong commitment to modern software development practices, performance optimization, and secure communication. It serves as a solid foundation for a full-scale real-time collaboration platform and demonstrates readiness for future growth and technological advancements in the evolving landscape of web-based communication systems

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